The same answer, two ways

Brute-force spin-up vs. the flux-anchored equilibrium solver

Why the old and new methods must reach the same periodic steady state — the argument, and the measurement.

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Claim. The fast flux-anchored solver (`lunar/equilibrium.py`) and a long brute-force spin-up are not two different models. They are two *routes to the same fixed point*: the unique periodic steady state of the regolith column. This note proves that — first by argument (the steady state is unique and both methods land on it), then by measurement (brute force, started from two very different temperatures, converges onto the shortcut's profile).

1 What "the answer" actually is

Both methods solve the same 1-D heat equation in the regolith,

$$\rho(z)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\left(K(T,z)\, \frac{\partial T}{\partial z} \right), \tag{1}$$

with the same boundary conditions: a non-linear radiative balance at the surface, and a fixed geothermal flux Q_b entering at the bottom. Driven by a periodic Sun, the column settles to a *periodic steady state* — a temperature field $T(z,t)$ that repeats every lunation. That settled state is "the answer" we want; the retrieval then asks which K_d makes it match the sensors.

The periodic steady state is pinned by two conditions:

1. **The surface skin is periodic.** In the top few skin depths the temperature swings through a repeating daily cycle set by the radiative balance.
2. **The deep column carries exactly Q_b.** Below the skin there are no heat sources, so in steady state the cycle-mean conductive flux is the *same at every depth* — and that value is the flux entering from below, Q_b. Writing the cycle mean as $\langle T \rangle$,

$$K(\langle T \rangle, z)\, \frac{d\langle T \rangle}{dz} = Q_b \qquad \Longrightarrow \qquad \frac{d\langle T \rangle}{dz} = \frac{Q_b}{K(\langle T \rangle, z)}. \tag{2}$$

In plain words

Condition 2 is just conservation of energy at steady state: with nothing adding or removing heat inside the rock, the same trickle of geothermal heat (Q_b) must pass through every layer. That *fixes how fast temperature rises with depth* — the gradient is Q_b divided by the conductivity. The deep temperature profile is therefore *not a free thing to be discovered*; it is determined by Eq. (2) once you know the temperature at any one depth.

2 The old way: brute-force spin-up

Start the column at some guessed temperature and integrate Eq. (1) forward, lunation after lunation, until it stops changing. This is correct in the limit of infinite time — but the catch is *how long* that takes. The column relaxes on the diffusive timescale of its full depth,

$$\tau \sim \frac{z_{\max}^2}{\kappa} \approx \text{ order } 10^3 \text{ lunations} \qquad (z_{\max} = 5\,\mathrm{m},\ \kappa \sim 10^{-8}\,\mathrm{m^2 s^{-1}}). \tag{3}$$

A fixed spin-up of a few tens of lunations is nowhere near that. At the sensor depths (0.8–2.4 m) the profile still *remembers the temperature it was started at.*

The old trap (audit flag F1)

Because the deep relaxation is so slow, the usual convergence check — "did $\langle T \rangle$ change by less than 0.01 K between successive lunations?" — *passes long before the column is actually settled.* The model looks converged while it is still drifting. The starting temperature then acts as a hidden free parameter, and it biases the retrieved K_d. That is the bug the new method removes: not by running longer, but by not needing to.

3 The new way: the flux-anchored shortcut

The shortcut does not wait for diffusion to *discover* Eq. (2) — it *imposes* it. One outer iteration is:

1. **Short spin-up (∼12 lunations).** Long enough to make the fast surface skin periodic against the current deep column, and to read the cycle-mean "anchor" temperature $\langle T \rangle(z_0)$ just below the skin ($z_0 \approx 0.55\,\mathrm{m}$). The skin is fast because its relaxation time is small, and its mean temperature is set by the surface balance — which barely cares about the deep column (Q_b is $\sim 10^{-4}$ of the diurnal flux scale).
2. **Reconstruct the deep profile.** From the anchor, integrate Eq. (2) downward to fill in $\langle T \rangle(z)$ for all $z > z_0$ — *instantly*, no time-stepping. (A small "rectified" eddy term $u_{\mathrm{rect}} = \langle K\,\partial_z T \rangle - K(\langle T \rangle)\,\partial_z \langle T \rangle$ is computed from the cycle and subtracted, so the closure stays exact near the skin.)
3. **Iterate.** Feed the reconstructed column back into another short spin-up and repeat. It converges in 3–5 rounds.

In plain words

Same destination, no waiting. Like finding a tank's steady water level from "inflow = outflow" instead of pouring water in and watching it slosh for an hour. The level is fixed by a balance; you can compute it directly.

4 Why they cannot disagree

This is the part worth being precise about.

1. **The steady state is unique.** Given the anchor value $\langle T \rangle(z_0)$, Eq. (2) is a first-order ODE with a smooth right-hand side, so it has *one* solution below the anchor. The anchor value itself is set by the surface energy balance, which the skin reaches quickly and essentially independently of the deep column. So there is exactly one periodic steady state.
2. **Brute force reaches it.** Equation (1) is a parabolic equation with periodic forcing. Its transient decays on the timescale τ, and what remains is the unique periodic response. So brute-force-to-convergence *is* that unique state — it just spends $\sim 10^3$ lunations getting there.
3. **The shortcut reaches it.** Its fixed point satisfies condition 1 (the skin is periodic, by step 1) *and* condition 2 (flux closure, by step 2) — the two conditions that define the steady state. So its fixed point is the same unique state.

The crux

The two methods enforce the *same* two conditions; the steady state satisfying them is *unique*. The only difference is bookkeeping: brute force discovers the deep gradient Q_b/K slowly, by letting heat diffuse; the shortcut writes it down directly from energy conservation. A different final answer would require a second steady state — and for this problem there isn't one. *They must agree.*

The one honest caveat: the reconstruction uses the mean-field form of Eq. (2) plus the rectification correction, so the shortcut equals the true steady state to a small, *controlled* tolerance (tens of mK at sensor depth, see below), not bit-for-bit. Brute force, in turn, only reaches that same tolerance after thousands of lunations. So "equal" means "equal to within a tolerance both methods can be pushed below," which is exactly what the measurement shows.

5 The proof: they land on the same profile

Convergence from a wrong start. Starting brute force at 240 K (the shortcut's settled value at 1 m is 250.38 K) and continuing, the gap closes monotonically and exponentially:

brute-force lunations	$\langle T \rangle$ at 1 m [K]	error vs. shortcut [K]
100	244.48	5.90
300	246.55	3.83
600	248.19	2.19
1000	249.36	1.02
shortcut	**250.38**	**(reference)**

The error roughly halves every few hundred lunations (e-folding ∼520 lunations) — a single exponential decaying onto the shortcut's value. Extrapolated, brute force matches the shortcut's certified tolerance at ∼3000 lunations.

Two starts, one curve. Run brute force all the way (3000 lunations) from *two* very different starts, 240 K and 260 K, and overlay both on the shortcut (Figure 1). The three profiles are indistinguishable across the whole 5 m column.

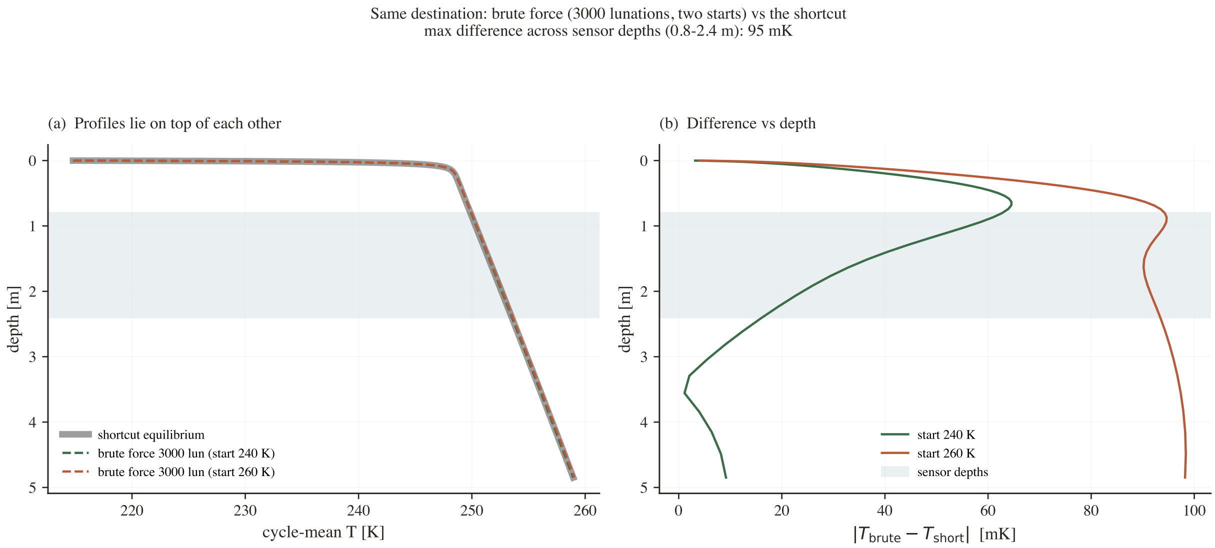

Figure 1: Brute force run to 3000 lunations from 240 K *and* 260 K (dashed), overlaid on the flux-anchored shortcut (solid grey). Left: the profiles lie on top of one another. Right: the residual is at most ∼60–95 mK across the sensor depths — and that remainder is simply brute force still finishing the last fraction of its slow descent (it keeps shrinking with more lunations). Two unrelated starting points and the direct method all reach one profile.

What we measured

Quantitatively. At 3000 lunations the fully-converged brute force agrees with the shortcut to **63 mK** (start 240 K) and **95 mK** (start 260 K) across the sensor depths (0.8–2.4 m) — and that gap is still closing. The shortcut reaches the same state in $\sim$9 s; brute force needs $\sim$4 min to match it.

The shortcut certifies its own convergence. It does not ask you to trust it. Every solve returns two diagnostics (Figure 2): the anchor *drift* between outer iterations (required < 0.03 K; typically $\sim$3 mK) and the *flux closure*, $\max |\langle q \rangle - Q_b|/Q_b$ below the anchor (required $< 2\%$). A direct guess-independence test — start the shortcut from 240 K and 260 K — gives sensor-depth temperatures agreeing to < 0.03 K. So the shortcut is provably at the steady state, which is precisely the check the old fixed-spin-up could not pass.

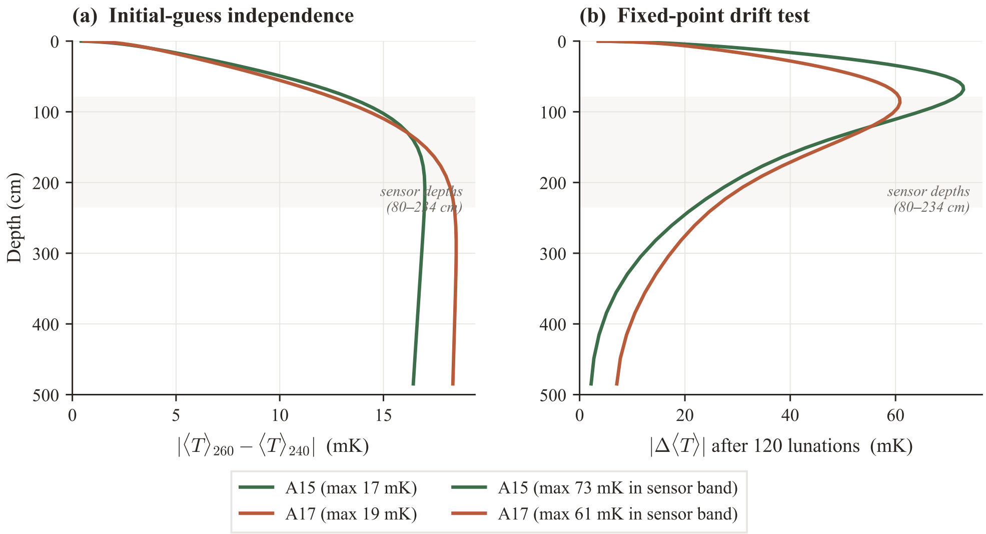

Figure 2: Solver certification: guess-independence and mean-flux closure on Q_b as functions of depth/iteration. These are the numbers that *prove* the shortcut arrived at the periodic steady state.

6 Bottom line

- **Same answer.** Both methods target the one periodic steady state. Brute force from any start converges onto the shortcut's profile (Figure 1); the argument in §4 says it has no choice.
- **$\sim$30× less work.** The shortcut reaches it in $\sim$9 s; brute force needs $\sim$3000 lunations ($\sim$4 min) per solve to match — and the retrieval needs hundreds of solves.
- **And it is certified.** `drift` and `flux_closure` prove convergence on every solve, removing the hidden-parameter bias (F1) that the old fixed-length spin-up silently carried.

Reproduce it yourself: `notebooks/equilibrium_demo.ipynb` runs both methods (the brute-force runs fan across CPU cores) and regenerates Figure 1; `pytest tests/test_equilibrium.py` runs the guess-independence and flux-closure checks.